

# Session 1: Transformer from First Principles

*Building a Mental Model of a Decoder-Only LM*

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*Training and Scaling Language Models: From First Principles to Efficient Serving*

# Lecture map

- ① Attention from first principles
- ② The decoder block
- ③ Evidence before scale

## Guiding question

What must happen between a token ID and a distribution over the next token?

# What should be clear by the end?

- 1 Track the shape and meaning of every tensor in causal self-attention.
- 2 Explain why scaling and masking happen before the softmax.
- 3 Trace information through a pre-norm decoder block.
- 4 Name the tests that make a minimal implementation trustworthy.

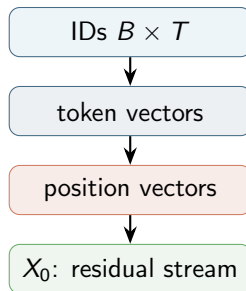
# Outline

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# Tokens become points in a shared space

- Input IDs have shape  $B \times T$ .
- A lookup table  $W_E \in \mathbb{R}^{V \times d}$  produces content vectors.
- Position information makes order visible to the model.

$$X_0 = W_E[\text{ids}] + P_{0:T} \in \mathbb{R}^{B \times T \times d}$$



# Attention is learned retrieval

## Query

What does this position need?

$$Q = XW_Q$$

## Key

What does each position offer?

$$K = XW_K$$

## Value

What content should move?

$$V = XW_V$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}} + M\right) V$$

## The distinction that matters

Queries and keys choose the mixture. Values supply the mixed information.

Source: Vaswani et al. and The Annotated Transformer

# Why divide by $\sqrt{d_h}$ ?

Assume independent query and key components with unit variance.

$$q^\top k = \sum_{i=1}^{d_h} q_i k_i \quad \text{Var}(q^\top k) \approx d_h$$

## Without scaling

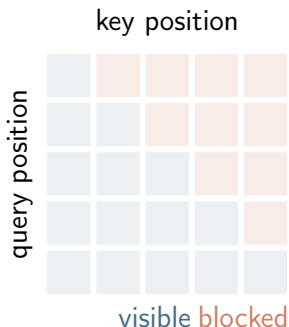
Wider heads create larger logits.  
Softmax saturates and gradients  
become poorly conditioned.

## With scaling

Dividing by  $\sqrt{d_h}$  keeps logit variance  
roughly constant across head widths.

**This is a variance-control argument, not a cosmetic constant.**

# Causality is a constraint on logits



- Position  $t$  may use keys  $0 \dots t$ .
- Future logits receive  $-\infty$  before softmax.
- Masking probabilities after softmax breaks normalization.

## Causality test

Change one future token. Earlier output logits must remain identical.

## Checkpoint: one query row

### Think, then compare

For sequence length four, write the four logits seen by the query at position two. Which entries survive the mask, and across which dimension must softmax normalize?

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# Multiple heads create multiple retrieval spaces

$$d_h = d/H$$

$$Q, K, V : B \times H \times T \times d_h$$

$$S = QK^T : B \times H \times T \times T$$

$$\text{concat}(O_h) : B \times T \times d$$

Heads run the same operation with different learned projections. The output projection mixes their results back into the residual stream.

## Shape invariant

$d$  must be divisible by  $H$ . A transpose mistake can preserve element count while changing semantics.

## Useful habit

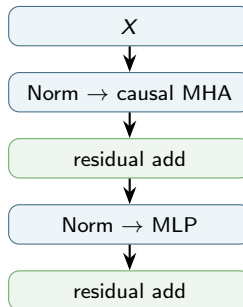
Annotate batch, head, query, key, and channel axes before every reshape.

# A pre-norm block has two residual updates

$$X' = X + \text{MHA}(\text{Norm}(X))$$

$$X'' = X' + \text{MLP}(\text{Norm}(X'))$$

- Attention mixes information across positions.
- The MLP transforms each position independently.
- Residual paths preserve a direct information route.



# Pre-norm and post-norm are different models

## Pre-norm

$$X + F(\text{Norm}(X))$$

The identity path bypasses normalization. Deep optimization is usually more forgiving.

## Post-norm

$$\text{Norm}(X + F(X))$$

This is the original Transformer form. It has a different gradient path and training behavior.

**Do not describe one convention and implement the other.**

# The language-model head turns states into a task

$$Z = \text{Norm}(X_N) W_{\text{vocab}}^{\top} \in \mathbb{R}^{B \times T \times V}$$

$$\mathcal{L} = -\frac{1}{BT} \sum_{b,t} \log p(x_{b,t+1} \mid x_{b,0:t})$$

Each position predicts the next token.  
The target is the input shifted by one position.

## One stream, two views

```
x = tokens[:, :-1]
```

```
y = tokens[:, 1:]
```

## Common bug

A wrong shift can produce a finite, decreasing loss while training the wrong task.

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# A shape ledger catches semantic bugs

| Tensor           | Shape                            | Meaning                 |
|------------------|----------------------------------|-------------------------|
| token IDs        | $B \times T$                     | discrete input          |
| residual stream  | $B \times T \times d$            | one state per position  |
| $Q, K, V$        | $B \times H \times T \times d_h$ | per-head projections    |
| attention scores | $B \times H \times T \times T$   | query-key compatibility |
| head output      | $B \times H \times T \times d_h$ | contextualized values   |
| logits           | $B \times T \times V$            | next-token scores       |

## Review question

Which axis does softmax consume, and which reshape reconstructs  $d$  without mixing tokens?

# Three tests establish a trustworthy baseline

- ① **Shape test:** every intermediate matches the ledger for several  $B$ ,  $T$ ,  $H$  values.
- ② **Causality test:** modifying a future token cannot change earlier logits.
- ③ **Tiny-batch overfit:** a small model drives loss near zero on one repeated batch.

## What each failure means

Shape failure suggests layout. Causality failure suggests masking. Overfit failure suggests optimization, targets, or the end-to-end data path.

# The same components support three model families

## Encoder-only

Bidirectional attention produces representations for understanding tasks.

## Encoder-decoder

A source encoder connects to a causal decoder through cross-attention.

## Decoder-only

One causal token stream defines next-token prediction and generation.

Modern LLMs vary attention type, position method, normalization, MLP, and sparsity while retaining this core residual architecture.

## Checkpoint: debugging order

### Think, then compare

A model runs and loss decreases slowly, but generation copies future tokens from the training sequence. Which test should have caught this, and which tensors would you inspect first?

# The durable ideas

- ① Attention performs content-addressed mixing through queries, keys, and values.
- ② Scaling stabilizes logits. Causal masking protects the prediction task.
- ③ A decoder block alternates cross-token mixing and per-token transformation.
- ④ Shapes, causality, and tiny-batch overfitting are the minimum evidence.